

Leakage-Free Multi-Site EV Charging Demand Forecasting: Cross-Site Generalization, Transfer Learning, and Grid-Aware Scheduling

Abstract

Precise short-term prediction of aggregate electric vehicle (EV) charging demand is robust in the areas of demand-side control, mitigation of the peak, and charging operations that are grid-aware. However, many published EV forecasting models cannot be compared or replicated, because of the utilization of single-site data, extra covariates or evaluation strategies that do not sustain time-varying sequence. A leakage-free time-aware forecasting model of the demand in a multi-site EV charging is suggested in this study on the basis of the ACN-Caltech data. The session level charge data of six charging stations are transformed into continuous hourly series of demand and all the models are trained by using only lagged historical demand characteristics, omitting calendar, weather, traffic, and price characteristics. Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost are assessed by such controlled environment with the strict chronological order of training, validation, and testing. Gradient-boosting models are more effective than simple baselines in within-site forecasting in the reported experiments with LightGBM having the lowest MAE (12.41) and CatBoost having the lowest RMSE (17.96). The paper then makes a comparison of cross-site generalization of the heterogeneous locations in charging, and suggests that the direct transfer between locations makes less accurate predictions, which suggests the effect of distribution shift between sites. In this aspect, a transfer-learning setting is researched and fine-tuning on target-site information results in the intended performance (MAE 9.256 and RMSE 13.783). Finally, the predictions are used illustratively in an experiment on the redistribution of loads in the context of scheduling to demonstrate how increased prediction of demand can be helpful in improving the reduction of the peaks as well as smoothing the loads. The main importance of the work is that it offers a leakage-free, reproducible and deployment oriented benchmark of EV charging demand forecasting which interrelates within-site prediction, cross-site adaptation and downstream deployment operational application.

Keywords: Electric vehicle charging demand forecasting; time-aware forecasting; cross-site generalization; transfer learning; grid-aware scheduling; load redistribution

1. Introduction

The electrification of road transport is accelerating rapidly, with electric vehicles (EVs) playing a central role in global decarbonization strategies. The International Energy Agency claims that the number of EVs is progressively increasing worldwide and in 2024, the electric cars are going to be sold in the amount approximated to 17 million vehicles, which is about a quarter of the newly sold passenger vehicles^[1]. This has been especially evident in China, Europe and the United States where extensive implementation of public and workplace charging facilities has been actively encouraged by national policy programs^[1, 2]. Although EVs are associated with large lifecycle greenhouse gas emission benefits over internal combustion engine vehicles^[1, 3], the rising penetration poses new operational issues to power distribution networks. The demand of EV charging has high temporal variation, randomized users, and clustering effect particularly in shared charging systems like work areas and open charging stations^[1, 4]. The inconsistent charging may greatly enhance peak demand, worsen deviations on voltage, and hasten the decline of the assets in low voltages^[5, 6]. It has been previously demonstrated that during working hours or in the early evening hours, concentrating the charging can result in overloading the transformers and in the areas of the local congestion unless managed in a proper way^[6]. This has led to making correct short-term projections of aggregate EV charging demand a key requirement of grid-sensitive scheduling, peak reduction, and deployment of demand-side flexibility^[5, 7]. Nevertheless, the demand of EV charging is a challenging issue by nature. The cost of charging is determined by non-homogenous and partially observable factors including the travel patterns of an individual, the arrival time, the length of the charging process, and user choice^[8, 9]. The empirical investigations continue to show the presence of both recurring patterns through time (e.g. daily and weekly) and non-recurring fluctuations which are motivated by the user behavior^[9, 10]. Such a combination of structured and stochastic dynamics makes EV charging demand fundamentally contrasting with the traditional residential or industrial loads, and much more challenging to predict, especially when only historical load observations are used. The literature on forecasting has suggested a great number of forecasting methods. Initial analyses were based on statistical time-series models like autoregressive integrated moving average (ARIMA) and exponential smoothing because they are easy to interpret and compute^[5, 9]. Although such models are able to model linear temporal dependencies, they tend to perform poorly when implemented

on nonlinear and highly varying patterns of charging in the real world^[10]. In order to overcome these drawbacks, machine learning techniques, such as support vector regression, decision trees, Random Forests, and gradient boosting, are popularly used to build non-linear regression in EV charging demand^[9, 10].

Deep learning models like long short-term memory (LSTM) networks, gated recurrent units (GRUs), convolutional neural networks (CNNs) and hybrid models have been shown to achieve better predictive performance in a number of studies in recent times^[9, 10]. Models based on transformers and attention have been also studied, especially when rich contextual features would be provided. Meanwhile, it has been demonstrated that ensemble learning methods involving more than one model are more robust by addressing the complementary demand patterns^[10]. Though these innovations have taken place, deep learning models usually demand high-quality and large volumes of data and features engineering which could not be attained in most real-world charging systems^[9, 10].

Although this has been made, the current literature has a number of substantial shortcomings. First, most research works use evaluation procedures that do not necessarily maintain a temporal order, e.g., random cross-validation or the use of overlapping train-test windows, which may cause temporal leakage and give overly optimistic performance estimates^[11, 12]. Second, most forecast techniques require exogenous variables like weather, traffic, electricity rate or calendar features. Such inputs are effective to enhance predictive performance, but they decrease predictive reproducibility and scale to data-constrained settings where these predictors might be unobservable or inaccurate^[11, 13]. Third, the vast majority of research and development is done based on single-site datasets, there is not much research on cross-site generalization between locations of charging where usage patterns are heterogeneous^[7]. This restrictiveness brings the necessity to a systematically structured, leakage-free, and repeatable forecasting model that assesses multi-site EV charging demand in severe chronological limitations with solely intrinsic historical load data^[11, 14]. In illustrative application, the forecasting models must be capable of acting on realistic information boundaries besides being resistant to different charging environments. Therefore, the analysis processes that keep the time limits and which do not rely on

the non-essential external features are preconditions of obtaining the credible approximations of the performance.

In addition, comparatively low focus has been put on cross-site generalization and transfer learning of EV charging demand forecasting especially in stringent time-conscious assessment frameworks^[15]. One should understand how the models developed on a specific location of charging can be applied to another location, and whether it is possible to apply adaptation to improve its performance on the new location. In solving these issues, this paper comes up with a time-sensitive and leakage-free aggregate EV charging demand prediction model based on the ACN-Caltech dataset^[14, 15]. The individual charging periods are converted to continuous hourly demand time series but the time structure of the charging process is retained. The supervised regression models are only trained on historical demand features that are lagged and calendar variables and external sources of data are not used. It is a design which separates the forecasting capability of historical demand as well as being applicable in scenarios where ancillary information is unavailable or unreliable^[8, 15]. Strict chronological training, validation and test splits are strictly used to compare the Linear Regression, Random Forest, XGBoost, LightGBM and CatBoost models within this controlled environment. In addition to within-site forecasting, the cross-site generalization is also taken into consideration during the study by testing the models of various charging facilities with different patterns of usage. To further elaborate on the deterioration of the performance on cross-site transfer, a situation of transfer-learning is assumed, where the models are trained with minimal target-site information.

Lastly, the obtained predictions are applied to illustrative scheduling-based redistribution of load experiment to indicate how the use of short-term demand forecasts can help reduce peaks and smooth the loading of EV charging processes^[16, 17]. The aspect of scheduling is not mentioned as an optimization problem, but it provides the impression of the operational viability of the adequate demand prediction taking into consideration the realistic constraints.

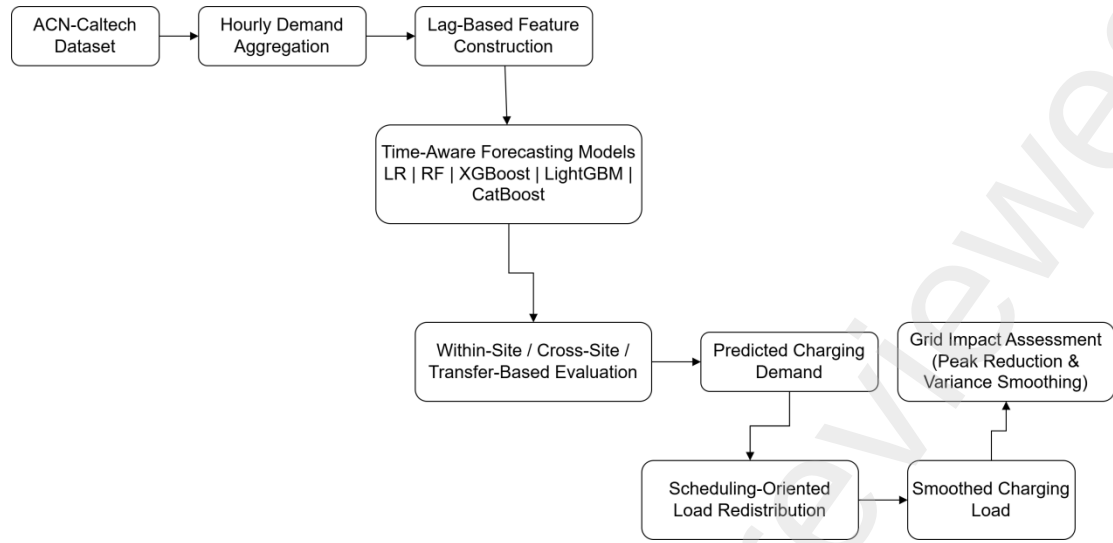


Fig. 1. Overall leakage-free forecasting, cross-site evaluation, transfer learning, and scheduling workflow.

This contribution can be regarded as providing to the effective and rigorous, reproducible, and deployment-friendly assessment model of EV charging demand forecasting. The main contributions are the following ones:

- 1) Time-sensitive leakage-free forecasting system, in which strict chronological distinction between training, validation and testing is maintained to allow realistic evaluation.
- 2) Comparison of machine learning model, using lagged historical demand characteristics, benchmark comparison of linear and tree-based models, and no external covariates are used.
- 3) Multi-site assessment of cross-site generalization, which brings out the effects of distribution shift between heterogeneous charging facilities.
- 4) A transfer-learning experiment that demonstrates that fine-tuning when limited amounts of target-site data are available can be employed to obtain superior predictive accuracy than when direct cross-site transfer is done.
- 5) One of the applications of forecasting in combination with load redistribution that is scheduling oriented and the way demand prediction can be applied to assist in alleviating peaks and smoothing of loads.

2. Dataset Description

2.1. Dataset Source

The current study relies on the ACN-Caltech Electric Vehicle Charging Dataset^[18] that is a log of real charging patterns of a group of EV charging stations. The single charging sessions are the representation of the data, and the charge delivered and the plug-in and unplug time are the characteristics of each of the single charging sessions. These session records are literally user charging behavior records and can be used to construct time indexed aggregated charging demand which can be used to make forecasts^[18].

Individual time series aggregated of the different charging stations in the ACN network including California Garage 01, California Garage 02, Arroyo Garage 01, North Wilson Garage 01, South Wilson Garage 01 and Parking Lot 01 are generated in the paper. The transformation pipeline of the same is applied to the facilities individually. This multi-site design can be used to test the forecasting models in a uniform manner under different charging conditions and cross-site generalization and transfer-learning experiments.

The data utilized in recent research is not more than the ACN-Caltech. There is no external data, synthetic data or auxiliary inputs. In particular, exogenous factors, i.e. weather conditions, electricity price, traffic statistics, or user-specific metadata, are purposefully omitted. The design further has the advantage that the forecasting structure can be entirely founded on inherent historical demand, and is therefore more reproducible and can be applied in data-constrained environments.

Table 2.1. Summary statistics of aggregated EV charging demand across charging facilities.

Garage	Number of Samples	Peak Load (kW)	Start Date	End Date	Time Resolution
Arroyo Garage 01	2409	177.41	1/1/2018	12/31/2020	1 hour
California Garage 01	16735	233.58	1/1/2018	12/31/2020	1 hour
California Garage 02	15035	206.33	1/1/2018	12/31/2020	1 hour
N Wilson Garage 01	7602	606.92	1/1/2018	12/31/2020	1 hour
Parking Lot 01	4410	41.01	1/1/2018	12/31/2020	1 hour
S Wilson Garage 01	8761	24.66	1/1/2018	12/31/2020	1 hour

2.2. Charging Sessions and Time Coverage

The mentioned amount of samples is related to non-zero aggregated hourly demand measurements that were not removed during preprocessing; thus, the numbers are lower than the actual total number of hourly timestamps within the 2018-2020 research timeframe. The data range is three years between January 1st, 2018 and December 31st, 2020, which is enough to ensure that seasonal and behavioral fluctuations in the demand of EV charging are covered. Hourly aggregate demand figures are obtained by adding up the charging demand of each time period. The time series obtained are considered chronological sequences and applied under severe time-conscious circumstances. There is no shuffling of any stage of data preparation. This maintains the time sequence and makes sure that no future knowledge is applied in the process of developing models. All datasets of a garage are worked on with the same chronological assumptions and aggregation process to allow fair comparison across the sites. The specific training, validation and testing procedure is as follows (Section 3).

2.3. Data Cleaning and Preprocessing

The preprocessing pipeline is aimed at maintaining the integrity in the time without artificial bias. The ordering of charging sessions is by the chronological order and then grouping and adding up all of the charging activity into hourly values as demand. Consistent principles are used during the preprocessing, they include:

- (1) Strict chronological ordering is maintained;
- (2) No random shuffling is applied;
- (3) Forecasts are generated using only past information;□

No explicit outlier values removal, missing value imputation or smoothing techniques are used, unless inherently required by the aggregation process. This conservative approach is deliberately used to ensure that no assumptions are made that may distort the underlying demand patterns or cause information leakage. Given that the dataset is modeled after actual operational measurements, by keeping the raw variability, the forecasting models can learn directly from actual noise and real fluctuations, which makes the models more robust and relevant for deployment.

2.3.1. Supervised Learning Framing for 24-Hour Forecasting

The forecasting problem is designed as a multi-output regression 24-hours-ahead problem. Considering historical aggregate demand data up to time t , this is the prediction of the model of following 24 hours of demand:

$$Y_t = [y_{t+1}, y_{t+2}, \dots, y_{t+24}] \quad (1)$$

In this case, the input variables are based on the lagged historical demand variables whereas the output vector is based on the future 24-hours of demand. The formulation allows one to directly forecast on a day-ahead basis, without the recursive multi-step prediction.

2.4. Descriptive Characteristics

This subsection gives descriptive summaries of aggregated EV charging demand to help interpret the forecasting results. The demand signals have very high temporal variation which implies that there is heterogeneous charging behavior as well as uneven arrival patterns across facilities. To define the temporal structure, intraday and weekly demand trends are analyzed, whereas the distribution of charging-session duration reveals further information about the variability of the usage. The aggregate demand patterns also vary among garages and this implies that there is location specific operational behavior and demand intensity.

The descriptive statistics in Fig.2. gives some background information on charging behavior and are useful in interpreting subsequent forecasting outcomes. Nonetheless, predictive performance within rigid time constraints is the main concern of this study as opposed to the large-scale discovery of session-level peculiarities.

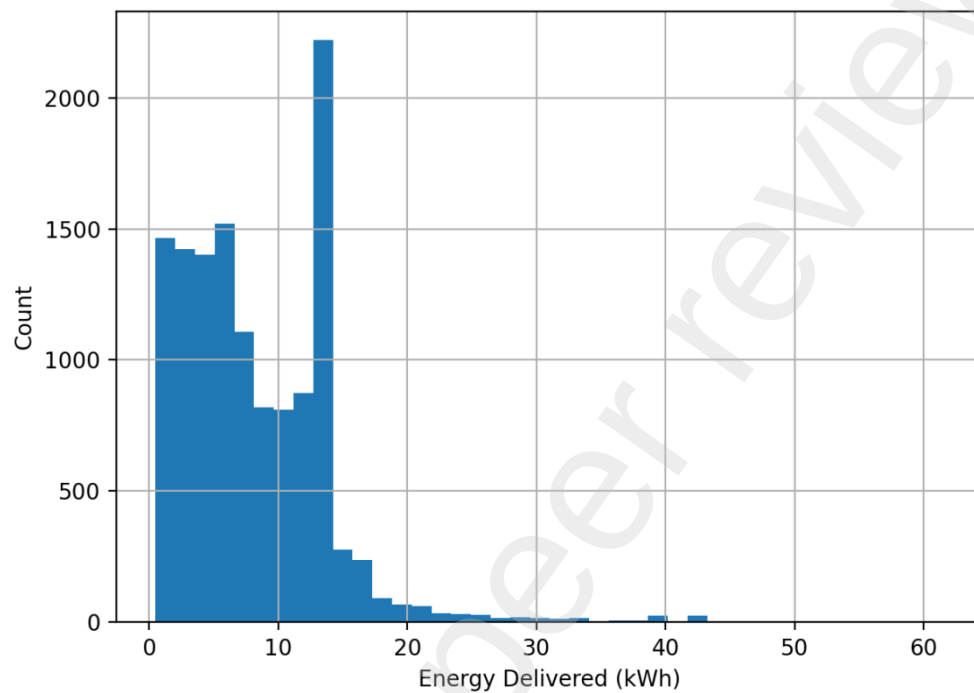


Fig. 2. Average EV charging demand by hour of day, showing intraday variation in charging activity.

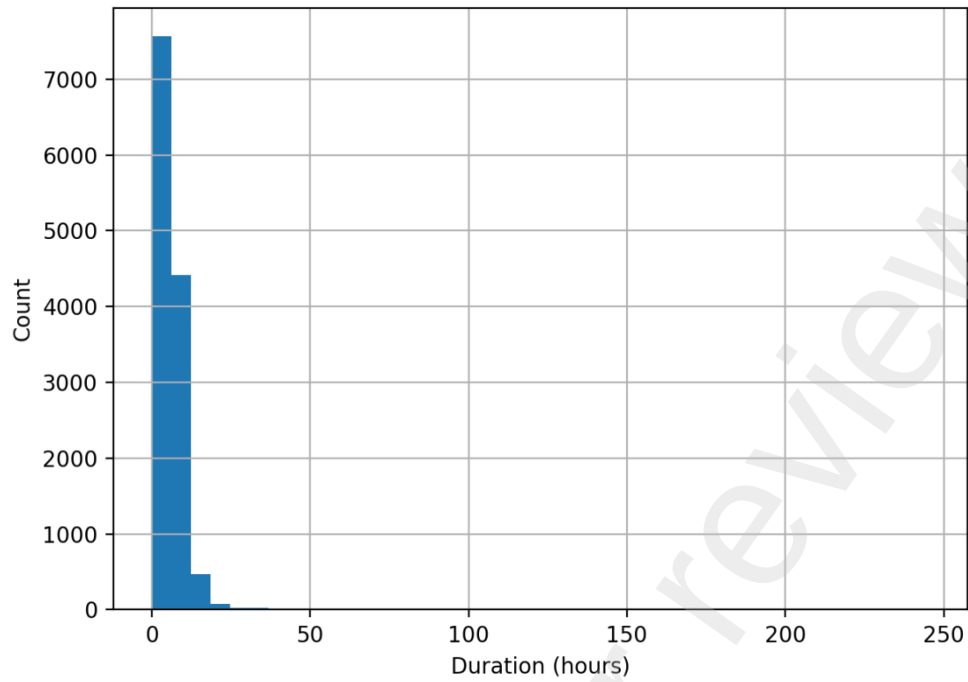


Fig. 3. Average EV charging demand by day of the week, showing variation in charging behavior across weekdays.

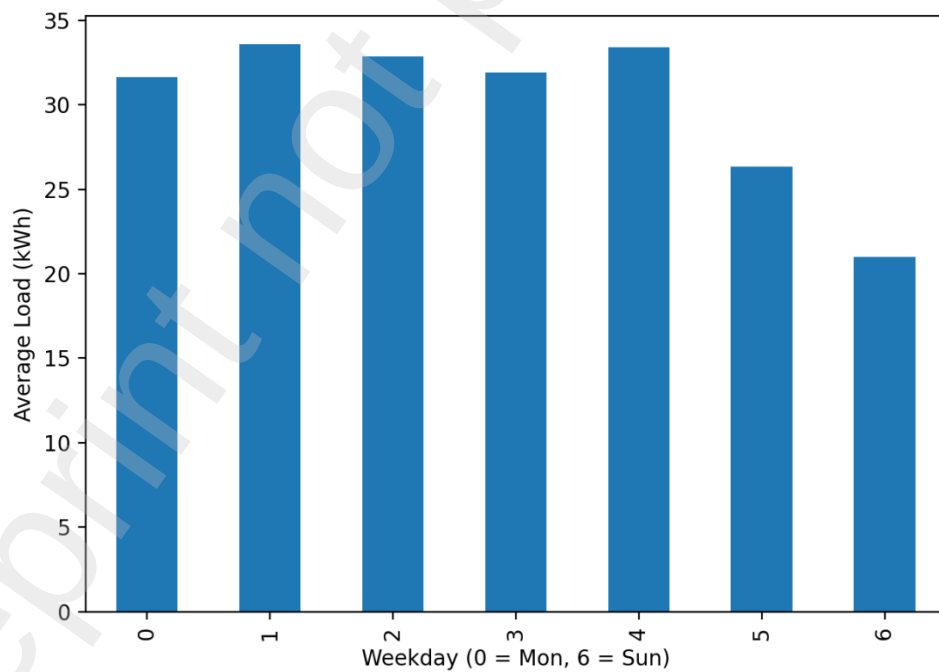


Fig. 4. Distribution of EV charging session durations across the study dataset.

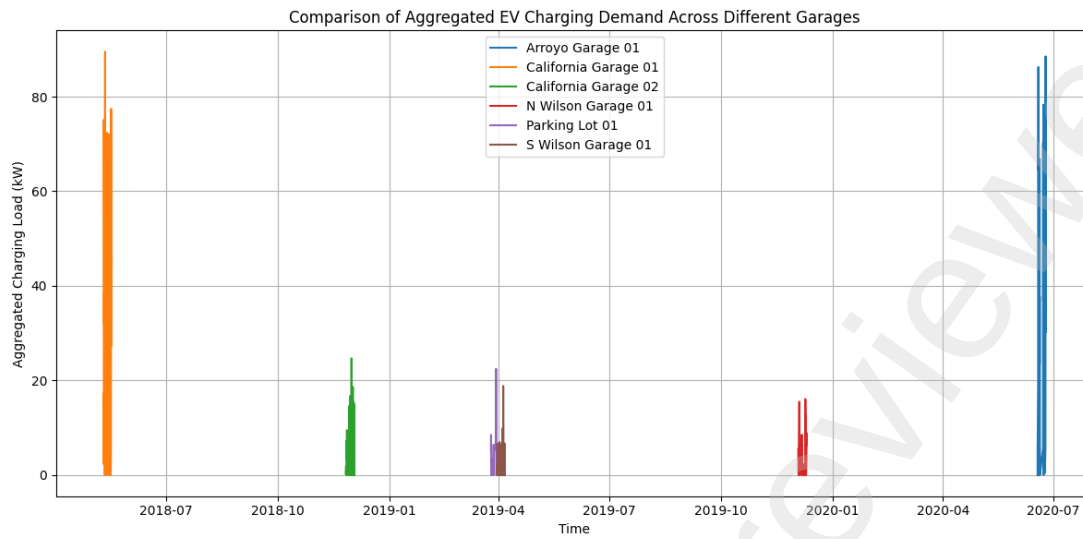


Fig. 5. Comparison of aggregated EV charging demand profiles across different garages.

2.5. Role of the Dataset in the Study

All the forecasting models are trained and tested by using only features obtained from historical charging demand. No external covariates or extra source of data is used. This allows for making sure that the evaluation is time-aware, e.g., that future information is not available during the training, yielding realistic and deployment-oriented performance estimates. The dataset was used in a strictly time-aware manner, such that the information from future observations was never available when model training and validation were performed. This strictly time-aware use of the dataset means that the forecast accuracy is assessed under realistic forward-looking conditions.

3. Methodology

3.1. Problem Formulation

This paper aims to make predictions of the aggregate electric vehicle (EV) charging demand with no other predictors except historical charging-demand data. Let y_t denote the EV Charging aggregated load at time step (t). The forecasting problem is to predict the future demand of horizon (h) by using the available information until time (t).

In case of general forecasting horizon, the prediction problem may be formulated as:

$$\hat{y}_{t+h} = f(\mathbf{x}_t) \quad (2)$$

The features vector based on lagged historical demand values at time t is denoted by $\mathbf{x}_t$. $f(\cdot)$ is the forecasting model $\hat{y}_{t+h}$ and is the estimated demand at horizon h . The key predictive environment in this experiment is a 24 hour ahead multi-output regression. Considering the lag-based features vector $\mathbf{x}_t$ the model will forecast the following 24 hourly demand values:

$$\hat{y}_{t+1:t+24} = f(\mathbf{x}_t) \quad (3)$$

Where;

$$\hat{y}_{t+1:t+24} = [\hat{y}_{t+1}, \hat{y}_{t+2}, \dots, \hat{y}_{t+24}] \quad (4)$$

This model allows making direct forecasting of aggregate EV charging demand without the need of recursive multi step prediction. The issue is considered to be a deterministic supervised regression problem. The present study does not encompass probabilistic forecasting, control via optimization and multi-objective formulations.

3.2. Time-Aware Forecasting Framework

Pure time-sensitive forecasting model is implemented to maintain chronological integrity during the preprocessing, model building and testing. At any particular stage, temporal ordering is preserved and no shuffling is done. This design eliminates the leakage of information of future observations to model training and gives a reasonable approximation of future deployment conditions. Allow the monitored dataset to be modeled as:

$$D = \{(\mathbf{x}_t, y_t)\}_{t=1}^T \quad (5)$$

where $\mathbf{x}_t$ is the lagged features vector obtained from the past observations, and y_t is the corresponding load value. In the experiments of cross-site generalization, the time series of the source and target garage are initially synchronized to the shared overlapping time period. Only past observations are then used in constructing lag

based samples. The forecasting model is trained on all the aligned source-garage data and tested on all the aligned target-garage data, each having an ordered source-target garage pair. This cross-site does not have a separate validation split.

The constraints are the following ones that are strictly followed:

- (1) no training is ever done with reference to future observations;
- (2) lag-based features are only attained using past values of loads; and
- (3) test performance is a measure of unobservable target-garage demand with prospective deployment.

This experiment has one forward transfer environment in order to compare controlled and repeatable garages. The current implementation does not make use of rolling-origin backtesting and repeated retraining. The train-validation-test data partitioning strategy is shown in Fig. 6. as chronological.

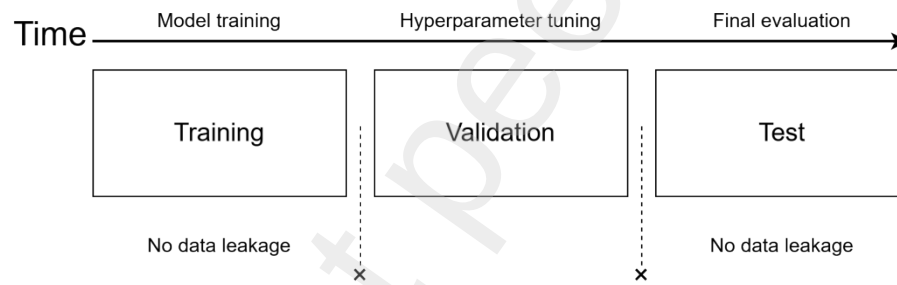


Fig. 6. Chronological source-garage training and target-garage testing strategy used in the cross-site experiments.

3.2.1. Lag-Based Features Construction

Each of the forecasting models is based on the lag-based feature representation built solely based on the past aggregate EV charging demand. The input vector $\mathbf{x}_t$ is developed on the basis of the last 24 hourly observations of the demand in the past at the point of the forecast.

In particular, the feature vector in terms of the lag can be defined as

$$\mathbf{x}_t = [y_{t-1}, y_{t-2}, y_{t-3}, \dots, y_{t-24}] \quad (6)$$

where y_{t-1} represents the cumulative EV charging demand at an hour i preceding time t . Therefore, the size of the lag window in the experiments of cross-site and transfer-learning is 24 hours.

All forecasting models are subjected to the same lag-feature construction process in order to have a fair and controlled comparison. This representation does not include no calendar variables, no weather variables, no traffic information, no electricity prices, or any other exogenous inputs. A schematic representation of the process of construction of features based on lag is depicted in Fig. 7.

3.3. Baseline Models

Within the framework of the same time-aware forecasting, several regression models are estimated with the help of a typical lag-based features representation. This is because the variation in performance is due to the modeling method and not the variation in input construction. The models that will be taken into account are Linear Regression, Random Forest, XGBoost, LightGBM, and CatBoost. Fig. 7 illustrates schematically the construction of the lag-based features.

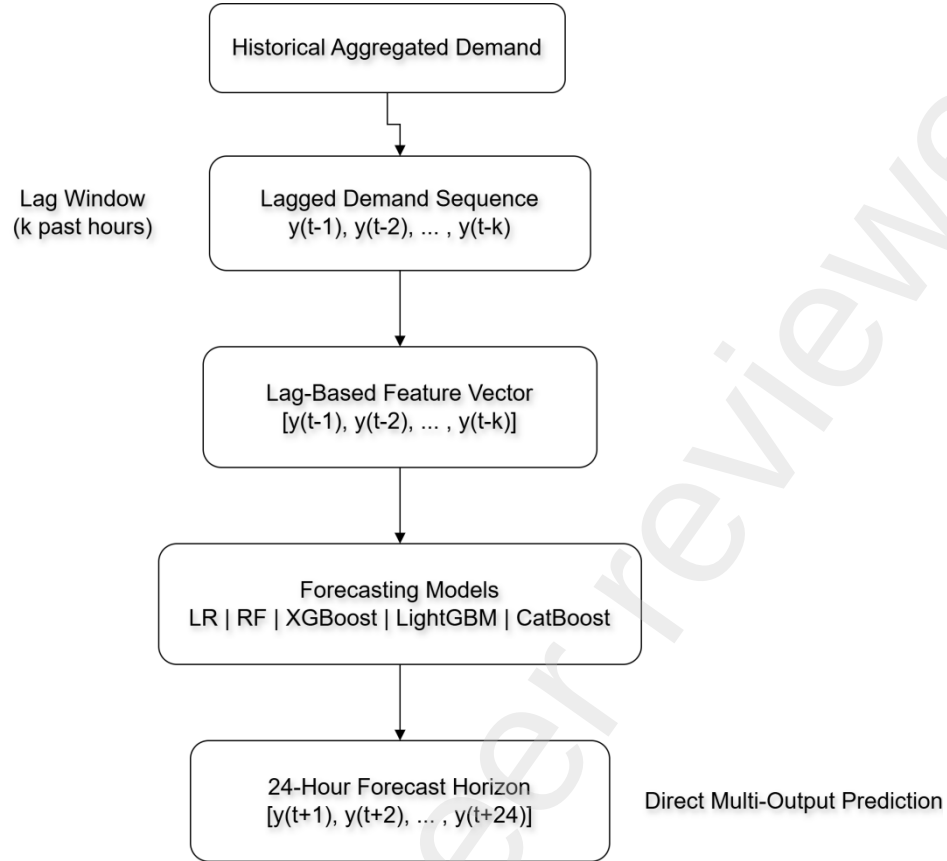


Fig. 7. Lag-based feature construction and multi-model forecasting pipeline for 24-hour EV charging demand prediction

3.3.1. Linear Regression

A simple baseline model that is applied to estimate the relation between aggregate EV charging demand and lagged demand features is the Linear Regression. The model presupposes that the target and the input vector are linearly related:

$$\hat{y}_t = \beta_0 + \sum_{i=1}^p \beta_i x_{t,i} \quad (7)$$

where β_0 is the intercept, β_i are regression coefficients, and $x_{t,i}$ denotes the i -th component of the lag-based features vector. No regularization techniques or fancy variants were used.

3.3.2. Random Forest

The nonlinear correlation between lag charging-demand variables and aggregate EV charging demand is modelled with the help of Random Forest regression. The model

is made up of a collection of decision trees that are trained using bootstrap samples of the training data, and the actual prediction is made by averaging the results of the constituent trees:

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^B f_b(\mathbf{x}_t) \quad (8)$$

Where B number of trees f_b prediction of the b -th decision tree

3.3.3. XGBoost

XGBoost is used as a gradient-boosted tree model that successively incorporates decision trees to minimise the residual errors of the present ensemble. The forecast would be in the form of:

$$\hat{y}_t = \sum_{k=1}^K f_k(\mathbf{x}_t), \quad f_k \in \mathcal{F} \quad (9)$$

Where K is total number of trees, $f_k(x_t)$ prediction of the k -th decision tree and $\mathcal{F}$ denotes the space of regression trees. XGBoost was learnt on the same lag-based representation of features and chronological protocol used on the other models in the cross-site experiments with fixed settings.

3.3.4. LightGBM

LightGBM is a gradient-boosting model that is efficient in structured data learning. It trains trees in the leaf-wise way, which may enhance training performance and predictive accuracy of tabular data. LightGBM in this paper is trained with the same protocol of lag-based features representation and in a chronological evaluation as with other models. Its predictive model is based on the general gradient-boosting of the form of Eq. (7).

3.3.5. CatBoost-Based Forecasting Model

The CatBoost is a gradient-boosted decision-tree model that is employed to predict aggregate EV charging demand. It is highly appropriate to tabular data that is organized and is able to represent nonlinear relationships among lagged demand characteristics and future charging load^[19]. Let $F_m(x)$ be the ensemble at the end of (m) iterations:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta g_m(\mathbf{x}), \quad m = 1, \dots, M \quad (10)$$

And the final prediction is

$$\hat{y}_t = F_M(\mathbf{x}_t)$$

Where $g_m(x)$ decision tree added in iteration m , η is learning rate and M total number of boosting rounds. CatBoost was applied in the cross-site experiments with fixed settings where the lag-based feature representation and chronological protocol were applied to the other models.

3.3.6. Model Settings and Hyperparameters

The cross-site experiments in the forecasting models were applied in a controlled and consistent environment so that they would be compared fairly. The default settings were applied to the Linear Regression. The application of the Random Forest was 100 trees and a constant random seed. XGBoost, LightGBM and CatBoost were also run with fixed random seeds and unspecified parameters were set to defaults of the library. Table 3.1 gives the exact model settings applied in the experiments.

Table 3.1. Model settings and hyperparameters used in the cross-site and transfer-learning experiments.

Model	Hyperparameters / Settings
Linear Regression	Default settings
Random Forest	n_estimators = 100, random_state = 42
XGBoost	verbosity = 0, random_state = 42
LightGBM	verbose = -1, random_state = 42
CatBoost (cross-site)	verbose = 0, random_state = 42
CatBoost (transfer learning)	iterations = 300, verbose = 0, random_state = 42

3.4. Training and Validation Strategy

The development of the models is based on a rigid chronological order. First, the training part of the site-specific time series is trained on every model. Second, the choice of hyperparameters is based on the performance on the next validation portion. Third, once the models have been selected the final set up is tested on the test held-out segment. There is neither random reshuffling, k-fold cross-validation nor rolling-origin retraining. All the models are trained and assessed on the same chronological partition and on the same lag-based features development process to obtain comparability. This design is controlled, and the effect of the choice of the model can be studied without variations in data management.

3.4.1. Within-Site Forecasting Protocol

The results of within-site forecasting are also provided to compare the results within the same lag-based demand representation. Here, training and evaluation of models are done with data of the same charging facility with constraints of time (chronologically). These findings are employed in the comparison of within site behavior versus cross site transfer performance.

3.4.2. Cross-Site Generalization Protocol

In order to measure whether the model is robust to distribution shift, cross-site generalization experiments were carried out by training forecasting models on data of one charging facility and evaluating them on a different unseen facility. In case of each ordered source-target garage pair, the source and target time series were aligned at their common time range of existence. The sample of lag based was then built with input features of the 24 hourly demand values which were obtained earlier. The training and testing of models were done on the whole aligned source-garage and the whole aligned target-garage datasets respectively. This cross-site did not have any separate validation split.

3.4.3. Transfer-Learning Protocol

It is believed that a transfer-learning environment can be used to improve the cross-site transfer, and a transfer-learning experiment was conducted using CatBoost. The

model pretrain was first run on California Garage 01 when input feature based on lagging was constructed using the most recent 24 hourly loads. After construction of lag-feature, 16, 711 useful training samples were obtained on the source-site dataset. The already trained model was then transferred to California Garage 02 and trained on the first 20% of the available target-site samples. This corresponded to 3003 samples of fine-tuning of the total number of 15011 samples of target-sites that were available after lag construction. The remainder 80 percent of target-sites which is 12,009 observations will be used in the testing out of the sample. Transfer was used as true fine-tuning whereby the already trained source-site CatBoost model was trained again on the target-site fine-tuning subset instead of being trained again in full. This was the procedure to ascertain whether the knowledge learnt in the source site might come in handy in the improvement of the forecasting accuracy with fewer data that was accessible on the target-site location.

3.5. Evaluation Metrics

In measuring accuracy of forecasting, mean Absolute Error (MAE) is used and root mean squared error (RMSE) is used. The metrics are calculated on out of sample predictions which are evaluated on the same conditions to all the models.

3.5.1. Mean Absolute Error (MAE)

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (11)$$

MAE is applied in calculating the average value of the prediction errors in original units of demand.

3.5.2. Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (12)$$

RMSE puts more emphasis on high prediction errors thus, this means that it is more sensitive to errors of peaks as compared to MAE.

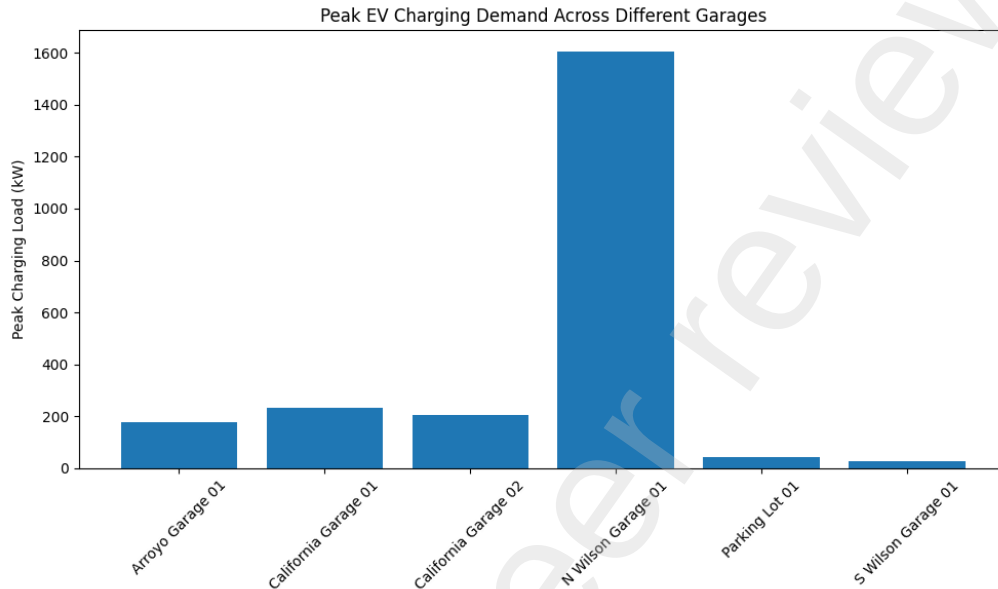
3.6. Scheduling-Oriented Load Redistribution

The time-based element of this paper is planned as a representative downstream implementation of short-term EV charging demand prediction. It aims at analyzing whether simple peak-oriented load redistribution analysis can be supported by the forecasted aggregate demand profiles. The forecasting model to be applied in this process is initially applied to produce anticipated charging-demand profile within the prediction horizon. In this profile, which is based on the forecasted profile, the candidate peak periods are the periods where the expected demand is relatively high, whereas the candidate redistribution periods are those where the expected demand is relatively low. A redistribution mechanism based on rules is then used to reallocate a portion of the projected load during the peak periods to the lower-demand periods in an effort to lower peak concentration and even out the aggregate load curve.

The redistribution process is not developed as an optimization problem, and does not explicitly represent charger-level scheduling constraints, uncertainty in the arrival of users, charging deadlines, constraints on network power-flow, and pricing goals. Rather, it is a simplified demonstration of a proof-of-concept of how the forecasting outputs can be linked with downstream operational analysis. The impact of redistribution is measured in terms of the difference between aggregate peak demand and load variance before and after the adjustment of the rule. These measures give an understandable signal of whether redistribution based on the forecasts can enhance the aggregate demand in the described experimental context.

Fig. 8. Peak EV Charging Demand Across Garages

4. Results



This part gives the timeframe and forecast results that were acquired in the strictly time-related frame as defined in the Section 3. All the experiments are performed using the ACN-Caltech dataset and lag-based historical demand characteristics only and no external input and covariate. These results are enumerated in the subsequent order (i) the temporal aspects of demand and forecast error behaviour, (ii) within-site forecasting, (iii) cross-site generalization and transfer learning, and (iv) the timing based experiment load redistribution.

4.1. Temporal Characteristics of Demand and Error Behavior.

EV charging demand follows evident time structure, which is intraday periodicity, weekday-specific variation, and significant inter-site variation. The demand in the chosen garages tends to be high in the daytime and low during the nighttime in the periods of low activity. Nevertheless, the scale and fluctuation of demand vary at different facilities due to different user behavior and disparity in the infrastructure use. Fig. 8 shows the typical demand patterns of the chosen garages in the form of examples and indicates common time patterns and local specifics. The comparison of the observed and predicted charging demand shows that the models reflect the overall structure of the demand signal at most operating ranges. The bigger the error in

prediction, the more such error is more likely to happen during the time of high demand or sudden short-term changes. This implies that even the overall predictability of forecasting is adequate; extreme demand events are still harder to predict than the more standard charging seasons.

4.2. Forecasting Performance Evaluation

The results shows in table 4.1 summary of the performance of forecasting in within-site and under the conditions of strict chronological training, validation and testing. The models that were evaluated are Linear Regression (LR), Random Forest (RF), XGBoost (XGB), LightGBM (LGBM) and CatBoost. The accuracy of the forecast is measured by the use of the Mean Absolute error (MAE) and the root mean squared error (RMSE).

Table 4.1. Forecasting performance comparison on the test set.

Model	MAE	RMSE
Linear Regression	20.85	26.14
Random Forest	16.32	20.21
XGBoost	12.65	18.38
LightGBM	12.41	18.15
CatBoost (Final)	12.42	17.96

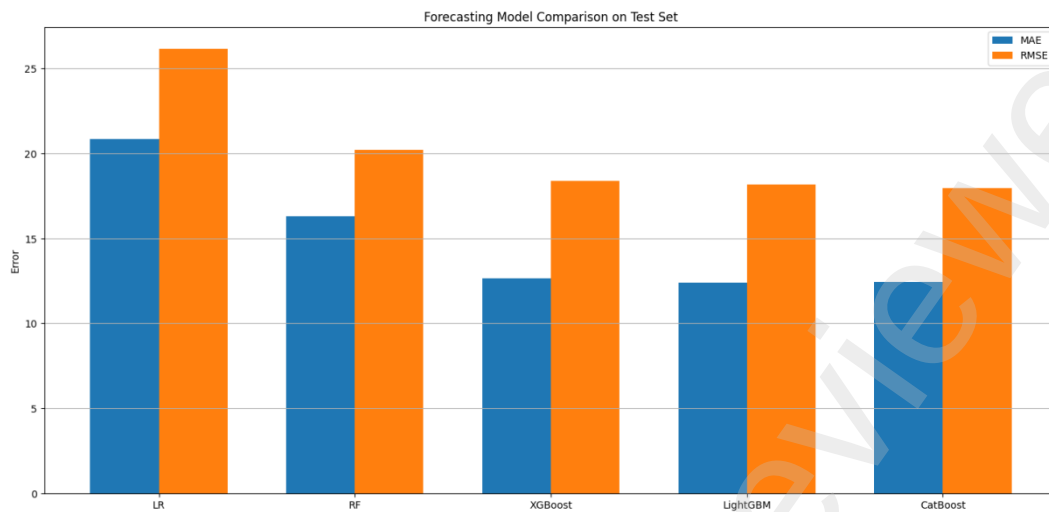


Fig. 9. Comparison of within-site forecasting performance in terms of MAE across the evaluated models.

According to Table 4.1, the linear model and the nonlinear tree-based models have a clear cut performance difference in within-site forecasting environment. Linear Regression (MAE = 20.85, RMSE = 26.14) has the best error, which is suggestive that a pure-linear mapping model is not only not sufficient to obtain the temporal variance of aggregate EV charging demand. On the contrary, the ensemble-based techniques obtain fewer errors and this means that nonlinear relationships in the lag-based demand model are applicable in predicting correctly. The models that have been evaluated have the lowest MAE and RMSE with LightGBM and CatBoost having 12.41 and 17.96, respectively. This difference shows that LightGBM has a slight higher average predictive power, but the CatBoost is slightly capable of holding larger prediction errors, which are more substantive to the RMSE. Nonetheless, the numerical difference between these two models is not robust, thus they cannot be deemed as the within-site setting strongly performers as opposed to being separated clearly. The selected CatBoost settings which was finalized to be selected has a validation MAE of 14.25 and a RMSE of 18.94 when the selection of the model was in progress. Our choice of the configuration is said to be reported by these validation values and no formal testing of statistical significance is conducted in this research.

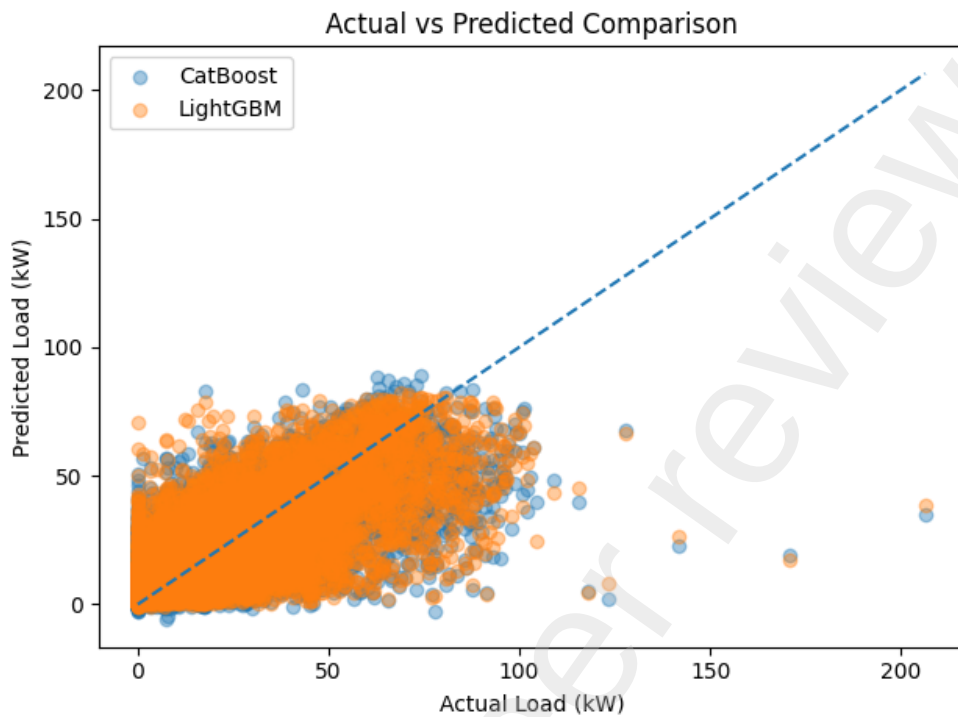


Fig. 10. Scatter comparison between observed and predicted EV charging demand for the LightGBM and CatBoost models.

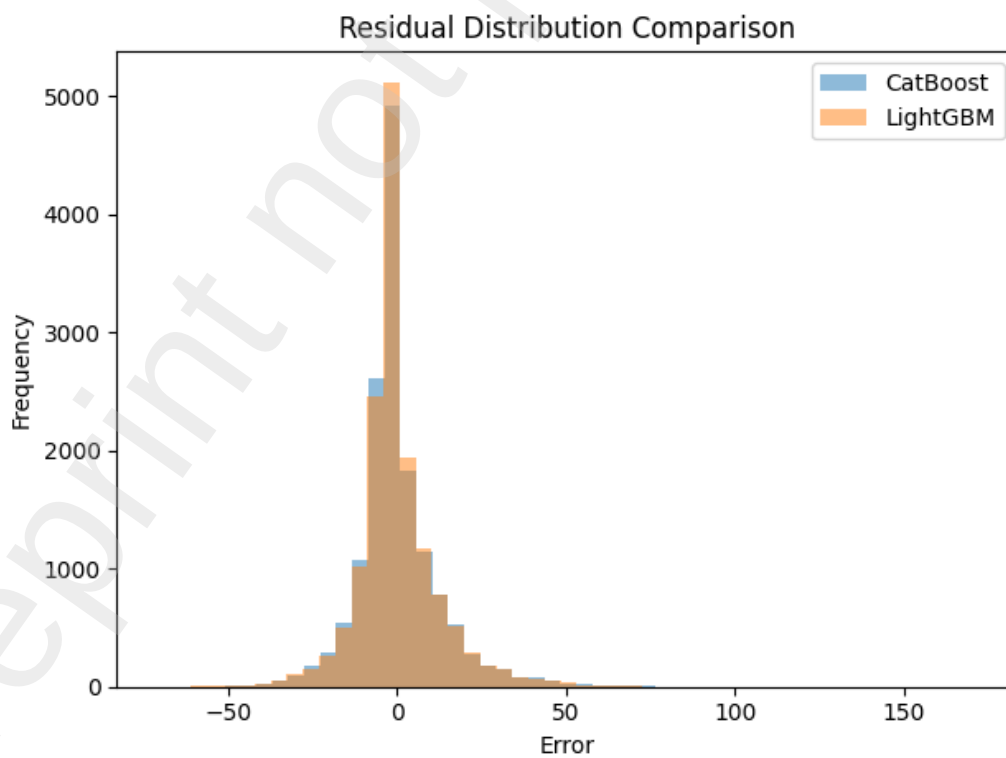


Fig. 11. Residual error distributions for the LightGBM and CatBoost models.

The above within-site results are achieved when models are trained and tested on data of the same charging facility when the data is split chronologically. In order to test the strongness of models in a heterogeneous charging environment, more cross-site experiments are done where models that were trained on one garage are tested on other unseen garages. Table 4.2 gives the average cross-site results.

Table 4.2. Cross-site generalization performance (average across garages)

Model	MAE	RMSE
Linear Regression	9.84	18.15
Random Forest	11.51	20.16
XGBoost	12.24	21.43
LightGBM	11.44	20.34
CatBoost	11.61	20.31

Observation: Table 4.2 indicates average cross-site performance in the settings of a garage-transfer. These values can be seen as a summary of another experimental protocol of the within-site findings in Table 4.1, which cannot be directly compared without corresponding per-site analysis.

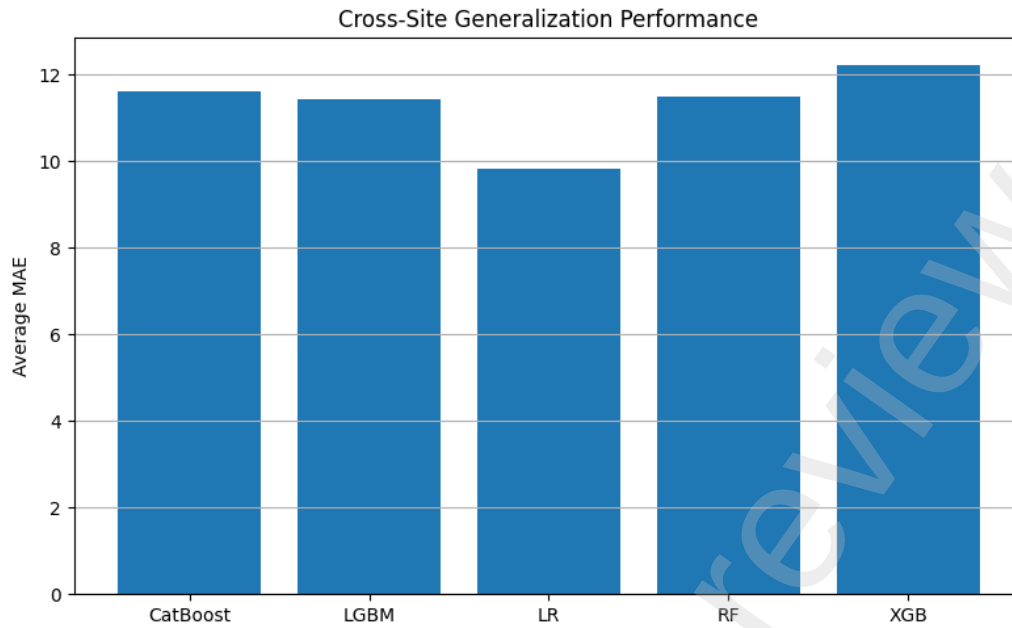


Fig. 12. Average cross-site forecasting performance across garages.

The cross-site findings show that model behavior varies in the case of transferring prediction between facilities with dissimilar demand characteristics. Linear Regression has the lowest MAE in the reported averages, and the tree-based models are no longer as advantageous as when within-site. This implies that the models that are favored by better nonlinear structure in the presence of within-site conditions might be more location-sensitive to location-specific demand patterns when directly relocated across garages. But Table 4.1 and Table 4.2 present the summary of various experimental settings and thus should not be taken as directly comparable without comparable per-site reporting or normalized evaluation. This is why, the cross-site analysis is applied to demonstrate the fact that model behavior varies when changing the site and the robustness across the heterogeneous charging facilities can be considered as the nontrivial problem.

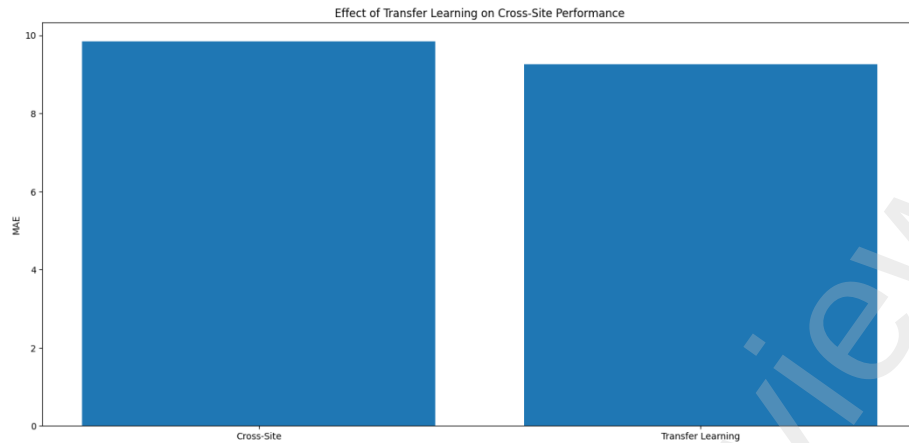


Fig. 13. Cross-site baseline and transfer-learning performance after fine-tuning on the target site.

In order to overcome the issue of cross-site generalization, the transfer-learning strategy was followed with CatBoost. The model was initially pretrained on California Garage 01 and then fine-tuned on California Garage 02 with little target-site data in this environment. In particular, the fine-tuning was performed on the first 20 percent of the available samples of the target-sites, whereas the rest of the 80 percent was to be used in testing. Following lag-feature construction, this equated to 3,002 fine-tuning samples and 12,009 test samples out of 15,011 target-sites that could be used. The results of the transfer-learning indicate a better forecasting performance compared to direct cross-site baseline in the given environment.

Table 4.3. Transfer-learning performance improvement after fine-tuning on California Garage 02.

Metric	Value
Cross-site MAE	11.6
Transfer MAE	9.256
Improvement (%)	20.2
Cross-site RMSE	16.9
Transfer RMSE	13.783

The reported results on the transfer-learning indicate better forecasting results compared to the reported direct cross-site baseline. After fine-tuning, MAE reduces to 9.256 as opposed to 11.60, and RMSE also reduces to 13.783 as opposed to 16.9. This is equivalent to a 20.2% improvement in MAE.

These findings demonstrate that knowledge acquired in the source site can enhance the accuracy of forecast in cases where pretrained model is then fine-tuned with limited target-site information. Transfer was used in this work as actual fine-tuning, where the previously trained CatBoost model was not retrained, but instead extended to the subset of target sites. The results hence endorse the usefulness of transfer learning in practice where there are limitations on labeled target-site data.

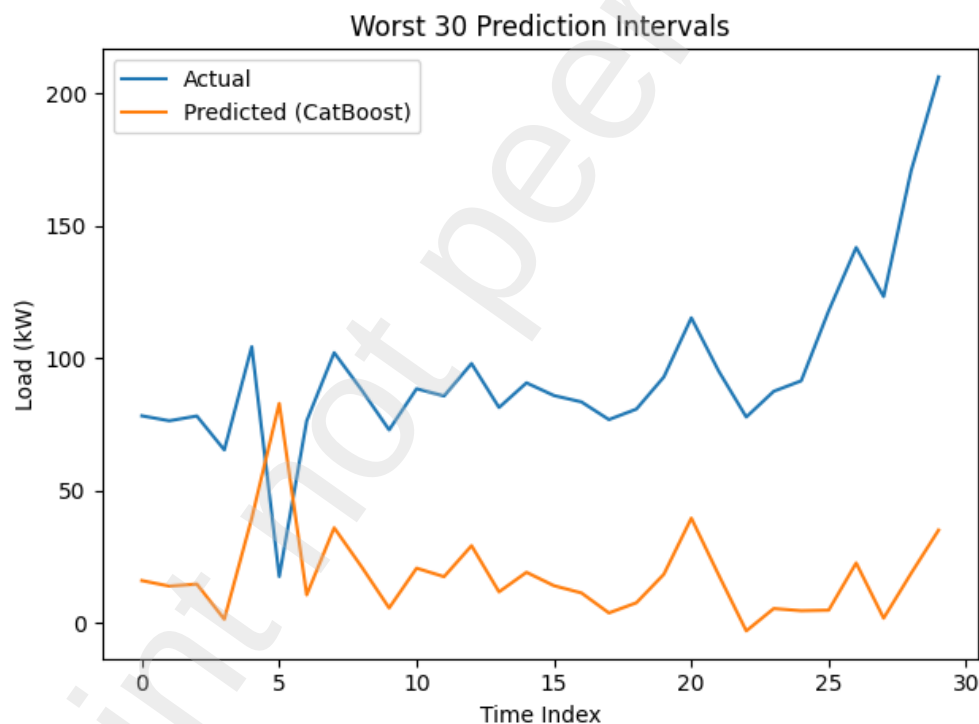


Fig. 14. Forecasting deviations during peak-demand periods.

4.3. Scheduling-Oriented Application Results

The scheduling-based application is premised on predicted EV charging request generated by the forecasting model outlined in Section 3. This experiment aims at redistribution of a portion of the predicted load during the high-demand times to the low-demand times to minimize peak concentration and flatten the aggregate demand

curve. The experiment is not developed as a comprehensive optimization-based scheduler; it is taken as an example of downstream application demonstrating how short-term demand predictions may be used to facilitate load-redistribution analysis in operations.

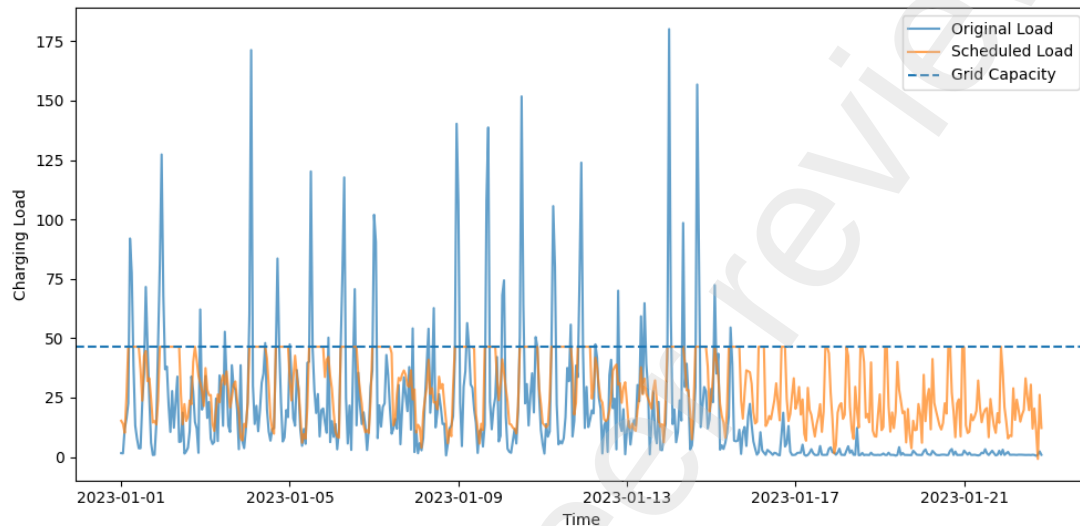


Fig. 15. Comparison of charging demand during high-load periods before and after scheduling based on forecasted demand.

Table 4.4. Scheduling-oriented load redistribution metrics before and after redistribution.

Metric	Before Scheduling	After Scheduling	Reduction (%)
Peak Load	180.12	46.40	74.24
Load Variance	704.72	194.44	72.41

The redistribution experiment in terms of scheduling generates substantial decreases in the peak demand and load variance compared with the pre-redistribution profile. Peak load decreases from 180.12 kW to 46.40 kW, while load variance decreases from 704.72 to 194.44. These findings suggest that redistribution under forecast

information can have a strong impact on the aggregate demand pattern of the reported experimental environment. Such gains must however be understood not as operationally definite but as illustrative. The redistribution system that is in place does not explicitly simulate charger-level constraints, user preferences, charging deadlines, network limits, and economic goals. Consequently, the mentioned gains must be regarded as a demonstration of the downstream potential instead of the proven deployable scheduling solution. Despite this weakness, the findings reveal that precise short-term prediction can be used to support peak-based demand management analysis in EV charging systems.

5. Discussion

Findings of the current research indicate that the forecasting of aggregate EV charging demand in short terms can be enhanced in case the assessment framework adheres to high chronological order. The tree-based ensemble models in the within-site context also have systematically lower forecasting error than the linear baseline, which means that there is nonlinear temporal structure in aggregate charging demand that is not well represented by the simple linear map involving past historical demand to future load^[20, 21]. LightGBM has the lowest MAE of all the models evaluated with CatBoost having the lowest RMSE, which implies that both models are well performing. Although forecasting the problem may be confined to historical demand alone, the load signal still has sufficient structure that nonlinear models can use to obtain valuable predictive data^[21, 24, 27].

These in-site findings are also significant due to the values of the reported errors, as well as due to the protocol employed to derive these values. This may be supported by the artificial performance of forecasting when the chronological integrity is not maintained in the course of preprocessing, feature construction, model development or evaluation^[23]. This risk is especially significant in EV charging demand forecasting as neighboring observations have a high correlation and demand patterns can be replicated on a hourly, daily and weekly basis^[22, 23]. When the future information is introduced in the process of learning by random shuffling or inappropriate splitting by time, the results that are reported are likely to overestimate model behavior during actual deployment conditions^[23]. By maintaining the time sequence during the working process and the fact that all the predictions are made based on the past only,

the current framework offers a more valid estimation of the future forecasting performance.

The residual behavior that is witnessed on the within-site setting gives further insight on the forecasting problem. The error in prediction tends to be around the zero direction most of the operating range, but tends to increase in the extreme deviations during the high demand periods or sudden short term fluctuations. Zhang et al. demonstrated that regularity in charging period is usually accompanied by regular usage pattern and recurring time schedules that can be learnt effectively only with lagged demand^[22]. In comparison, peak-demand periods are also more likely to be indicative of concentrated arrival patterns, variable charging time, scarcity of chargers and other types of stochastic user behavior^[25, 26]. These periods are more challenging to forecast by nature especially when the model is purposefully narrowed down to historic aggregate demand as opposed to more contextual rich features^[22, 28]. The fact that error increases in peak periods thus indicates modeling challenge as well as complexity of the EV charging systems.

The practical relevance is secondly provided by the cross-site results. A model that has been trained on a single charging facility in the real deployment can require transfer to a different location where there is a lack of local data^[22, 25]. The averages reported indicate that model behavior is not the same as the one in cases where prediction is transferred between garages of different demand characteristics. The nonlinear tree based models that had been found to be advantageous in the within-site setting are no longer maintained as such in the direct cross-site transfer. This implies that aggregate EV charging demand is highly site-dependent and even that facilities have differences that cannot be accounted for entirely using a common historical-demand model^[22, 25, 29]. The cross-site results hence point to a significant deployment issue that good performance at one location would not necessarily equate to good performance at another.

This result is given an important extension by the transfer-learning experiment. In spite of the limitation of direct cross-site generalization, the results of fine-tuning that have been reported indicate that predictive performance can be enhanced in cases where knowledge that is gained in a source site is integrated with minimal data on the

target site^[21, 22]. This implies that there is certain commonality between the structures of the facilities, although the shared information will not be adequate to facilitate proper forecasting without local modification. Charging facilities or facilities with a short observation history may lack sufficient target-specific data to learn good models that have been trained from scratch, but may still learn good models that are pretrained on similar facilities and then fine-tuned on the few local observations^[21, 22].

The other advantage of the framework is the fact that it only uses lagged historical demand features. Most EV charging forecasting research works include exogenous factors, including weather, traffic, electricity costs, or specific calendar features^[20, 22, 24]. These variables can be useful in enhancing prediction accuracy in certain contexts but they minimize reproducibility and can make it difficult to deploy in contexts where such signals are not always available and consistent between sites^[20, 22]. This is because the analysis leaves the predictive value of the intrinsic historical demand alone by eliminating exogenous inputs in the current research, and maintains the comparison across the models interpretable. This design option will enable the research to find an answer to a narrow question: to what extent can forecasting performance be enhanced with the help of purely historical aggregate demand in the conditions of leakage-free and time-consciousness^[23]?

The experiment of scheduling-oriented load redistribution allows to expand the applicability of the forecasting results not only on prediction. The redistribution based on the forecast in the given case study results in significant decreases in maximum demand and load variation in comparison to the initial load profile. This implies that downstream effects of forecasting accuracy can be made on peak management and load smoothing^[26, 27]. Meanwhile, this part of the research should be taken into consideration. It is a simplified redistribution structure that is not meant to be a deployable charging controller. It does not include charger-level constraints, user preferences, departure constraints, queuing influences, network power-flow constraints and economic goals. In this regard, it makes sense that the value of this component is depicted as operational relevance instead of purporting a full-scale solution in terms of scheduling^[27].

There are also several limitations that are to be considered. It is analyzed on the foundation of the ACN-Caltech data that demonstrates the workplace-based charging facilities in a particular institutional and geographical setting^[25]. The demand patterns in this case might not be similar to the residential charging, public fast-charging corridors or mixed-use urban networks. Moreover, even though the dataset covers several years, the present analysis is not yet capable of fully disaggregating the analysis of seasonal robustness, horizon-wise variation of performance or per-site transfer sensitivity of all locations. The careful omission of exogenous variables enhances comparability and reproducibility at the cost of potentially restricted accuracy in future projections in the circumstances when there are good external signals^[25-27, 29]. The future work can expand the framework by test more datasets of public EV charging, adding chosen exogenous variables by the same strict time-aware protocol, and comparing advanced deep-learning structures within the same evaluation framework, and combining forecasting as more closely related to optimization-based control^[27, 29]. In general, the given research is a deployment-focused benchmark of EV charging demand forecasting and demonstrates that reliable assessment, awareness of the location, and the utility of the downstream are all the necessary conditions of viable advancement.

6. Conclusions

The paper provides the leakage-free and repeatable model of short-term prediction of aggregate EV charging demand based on the real-world data of the ACN-Caltech network^[25]. The model merely uses lagged historical demand characteristics and uses harsh chronological assessment in order to give pragmatic futuristic evaluation. Besides prediction, the work contains a load redistribution experiment that is based on scheduling to demonstrate how prediction of the demand in the short-term can be used to help with the downstream operations analysis.

The within-site setting indicates that the tree based ensemble models have lower forecasting error compared to the linear baseline in similar experiment conditions. LightGBM has the lowest MAE, and CatBoost has the lowest RMSE, pointing to the fact that both models work well in the described environment. The above results indicate that the historical aggregate demand has enough information to allow

efficient short term forecasting in case of evaluated nonlinear models on a leakage free protocol.

The cross-site experiments indicate that the performance can be forecasted on the changes in the performance after models are moved between the heterogeneous charging facilities. This implies that good performance in one location does not necessarily mean that the performance will equally be good in another, and that EV charging demand still has significant site-structure. This observation is generalized in the transfer-learning experiment. CatBoost model pretrained on California Garage 01 and fine-tuned on small California Garage data set yields a better forecasting performance as compared to the direct cross-site baseline, indicating that in cases where target-site data are scarce a path to a more effective forecasting can be offered by fine-tuning.

The experiment with the scheduling one generates significant decreases in the peak demand and load variance in the case study reported. These results are, however, to be considered with some caution since the redistribution process is modeled as a simplified heuristic on the aggregate level and fails to take into consideration charger-level capacity, user behavior, network capacity, and optimisation goals. Based on this, the scheduling component can be perceived as a pictorial downstream application and not a deployable scheduling solution.

The primary value of the work is a rigorous, reproducible and deployment-ready assessment system of EV charging demand forecasting in the face of realistic information assumptions. There are also limitations in the study: it utilizes one single public dataset representing the facilities of workplace charging, the exogenous variables are not included in order to maintain the comparability, a simplified demonstration of the scheduling is utilized. Thus, the presented results are to be taken as a piece of evidence in the context of the study and not as the conclusions that are to be applied to all charging environments, all the settings of operation, and the cases of deployment in the future. The possible future studies can expand on the framework based on assessing more datasets, integrating the chosen external variables within a rigid time-conscious set of circumstances, experimenting with more sophisticated

deep-learning methods, and investigating the idea of probabilistic forecasting and control based on optimization.

Data Availability Statement:

The data to be utilized in this research is grounded on the publicly accessible ACN-Caltech Electric Vehicle Charging Dataset. The authors are able to make the processed data and implementation details used in the experiments available on reasonable request.

Code Availability Statement:

The preprocessing code of the dataset, the training of the forecasting models, and the provided results are reported, and the code to do so is also accessible with the respective author with reasonable request.

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Competing Interests are declared:

The authors indicate that they do not have any known conflicting financial interests or personal relationships that may have emerged in the work they report in this paper.

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